# Eliciting latent knowledge: an epistemically honest “SmartVault”

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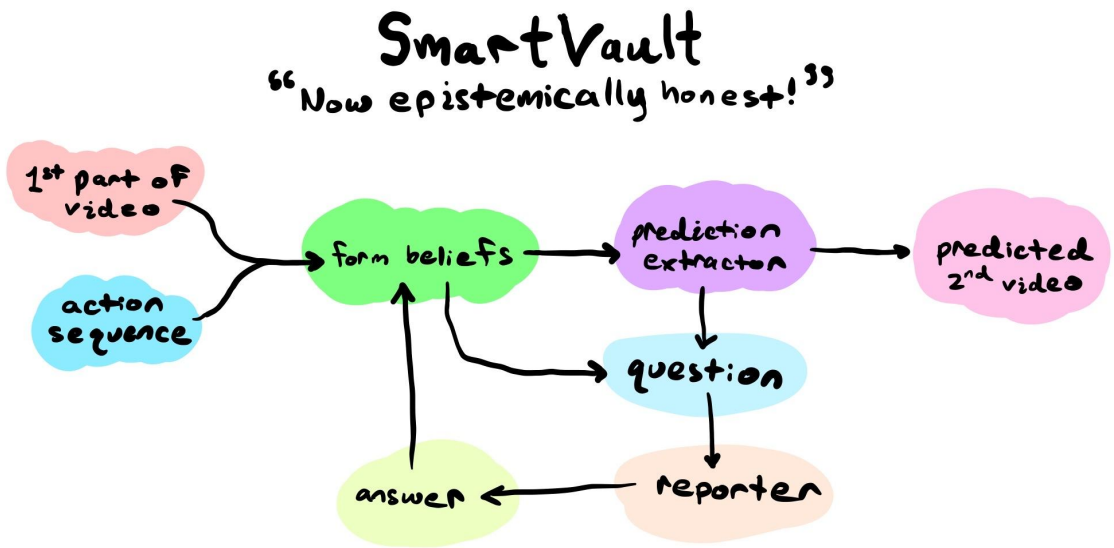
## Basic training strategy and why it might work

The central issue with eliciting latent knowledge in the SmartVault (SV) example is that an SV’s reporter can lie to a human inquisitor (HI) without losing effectiveness in the predictor; **our proposed solution is to force the predictor to adopt beliefs as stated by the reporter.** When the reporter misrepresents some domain-relevant fact or inference, the predictor temporarily adopts this belief and is expected to incur penalties through its loss function with the deficit being passed to the reporter. We adapt the SV chart so as to illustrate an “epistemically honest SmartVault” (ehSV).




We will assume throughout that the reporter has read and write access to the predictor’s Bayes net $B_{pr}$ as well as an HI’s Bayes net $B_h$.[1] A unit of training / test datum has

- Video and action sequence data
- An encoding of what the HI believes in the scenario[2]
- Questions about the training scenario

and is labeled with the valid video prediction and some indicator of how accurate the HI’s beliefs were. We train the predictor on the labeled training data through gradient descent on a loss function $L_{pr}$.[3] Likewise, we train the reporter to construct a dictionary between $B_h$ and $B_{pr}$ by performing gradient descent on a loss function $L_{rep}$ which, crucially, depends on input from $L_{pr}$.

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[1] We think we could task the ehSV with constructing a HI’s Bayes net during training by giving it labeled data as discussed on pp.25-7 of the ELK report. However we’ll suppress this for now and simply assume that the reporter has read and write access to the HI's knowledge and beliefs, unambiguously encoded in the Bayes net.

[2] Some configuration of priors in $B$ .

[3] One might also initially train the predictor on the human prediction data in some subset where humans make good predictions before moving onto the full dataset. This might have the advantage of increasing the speed of constructing the predictor as well as improving the likelihood of good translatability between the human and predictor Bayes nets. In later stages, the predictor should either ignore or discount human beliefs.

The reporter answers questions by translating portions of the graph underlying $B_h$ into computations on corresponding subnets of $B_{pr}$. These are translated back by the reporter which updates priors where applicable in $B_h$.[4] Finally, the reporter passes the newly updated $B_h$ into a function that updates $B_{pr}$. This translation of a translation of the optimized prediction network is then either accepted and subjected to prediction tests under its re-written priors or rejected. This whole process is analogous to a student posing a question to their teacher, the teacher responds with an answer and the student replies with a synthesis of the teacher's explanation, to which the teacher either accepts the student's synthesis and demonstrates that the synthesized version of the student's understanding of the teacher's explanation is not far off from the truth by, say, corroborating it with a Wikipedia article or correct the student and / or attempt to explain it differently. When the predictor finds an effective prediction strategy and the HI poses a domain-relevant question, the entangled loss functions should disincentivize the reporter from lying under threat of worsening the predictor's strategy. Barring issues with implementation, the ehSV directly avoids the following counterexamples:

1. Inference from $B_h$ is easier than inference from $B_{pr}$ (pp.41-2) and "Human belief understood before predictor belief" (p.44)
    a. Whether human belief is easier to understand or is understood before predictor belief is irrelevant if the reporter/predictor understands the associated belief makes worse predictions.
2. Memorizing predictors and answering nonsensically to unexpected inputs (pp.46-7)
    a. It seems safe to say that "nonsensical answers" would arise from "noisey" configurations of either the translated versions of $B_h$ or $B_{pr}$ and could be expected to yield bad predictions.
3. Bad reporter plays "look consistent game" (p.49)
    a. If the described implementation is possible, there is no "pretending to consistency" as posed in the original report: the predictor performs badly when the reporter lies or misunderstands questions.

We acknowledge that ehSV can likely hold some bad beliefs without significantly changing prediction quality and discuss this in more detail in a counterexample. We also remark that the strategies "penalize complexity" (pp.38-39) and "penalize computation time" (pp.40-1) seem like promising resolutions since the structure of the reporter in the ehSV seems as if it should avoid their respective counterexamples as posed in the ELK report: the efficiency gains from computations on small $B_h$ can be paltry relative to the cost of inserting bad prediction strategies into $B_{pr}$.

## More precise detail on strategy

We encode a Bayes net $B_*$ as a struct with two fields:

$$\beta_* := \textit{vector encoding priors}$$

$$\Gamma(B_*) := \textit{a Boolean incidence matrix encoding the underlying graph}$$

[4] We concede and intentionally suppress the possibility the reporter may need to enlarge the base graph or otherwise restructure *B* to faithfully convey the predictor's beliefs.

For us, “priors” for a node are all primitive and conditional probabilities encoding dependencies in the Bayes net. We write $A_{ij} \subseteq B_{ij}$ when $A_{ij}$ is a subgraph of $B_{ij}$. In our realization, this means $\neg(A_{ij}) \vee B_{ij}$ is true across all $i, j$. We note that a union of subgraphs can be found with a coordinate-wise sum of their boolean matrices using the “or” operation. To get a complement of a graph, we can do coordinate-wise “nand” operation between its matrix and that of the ambient graph.

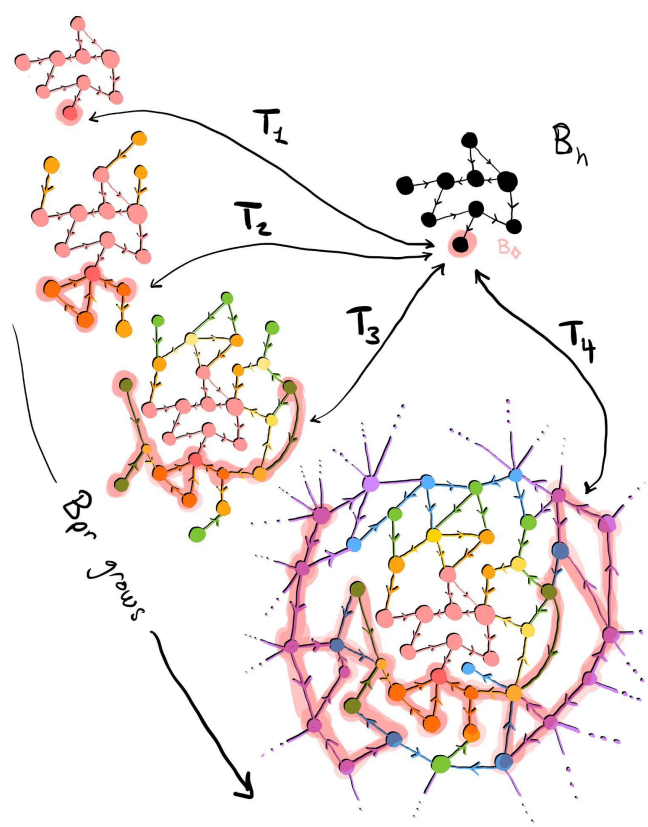
## Asking questions

For us, questions are subgraphs of $\Gamma(B_h)$.[5] For instance, the NL question “Is the diamond in the room?” can be parsed as “When the reporter finds the corresponding subgraph $\Gamma(B_\diamond) \subset \Gamma(B_h)$ encoding knowledge of the diamond’s position into $B_{pr}$ and the predictor performs the Bayesian computation associated to the translation of $\Gamma(B_\diamond)$, what’s the latter’s determination?”

## Leaping across ontologies with persistence features

We make the simplifying assumption that the predictor net first learns $B_h$ in the early stages of training where humans make good predictions, we have a starting point for mutual intelligibility as the predictor model grows by allowing us to track feature persistence in a manner inspired by persistent homology. Let $B_\diamond \subset B_h$ correspond to the NL question “Where is the diamond?”

Once $B_{pr}$ has learned $B_h$, this subnet should translate over in some trivial way under one of the early translation maps. Tracking the growth of $B_{pr}$ and, in particular, when new subnets connect with earlier subnets previously established to correspond with $B_\diamond$ allows us to collapse large portions of $B_{pr}$ into a persistence feature which is a “reasonable translation” of $B_\diamond$ into the predictor’s ontology.[6] In the image below: we see that as $B_{pr}$ grows, it expands upon the older subnet corresponding to an earlier translation of $B_\diamond$. Tracking the new nodes that feed into the subnet established as “conceptually close to $B_\diamond$” offers a coherent notion of a translation of $B_\diamond$ into some high-level feature of $B_{pr}$. We think this offers a route for resolving ontological mismatch.




[5] Since we’re assuming $B$ with the priors encodes the totality of the H that any domain-relevant question can be encoded as a subgraph of $B$

[6] We note that persistent homology would be useful for determining how conceptually close a subnet is to the high-level concept as expressed in $B$ since conceptually-relevant subnets seem as if they should fold into an established translation of B◇ earlier than those that are conceptually distant.

## Holding the reporter accountable

Our reporter consists of several maps with read / write privileges on copies of $B_h$ and $B_{pr}$. To prevent collaborative bad behavior, we conceive at least two of these maps as having an adversarial relationship which we hope establishes mutual intelligibility between $B_h$ and $B_{pr}$.

We think our reporter needs at least three maps:

- A map $T$ of graphs governed by hidden variables $t$ yielding the learned correspondence between concepts encoded in $B_h$ and concepts in $B_{pr}$.
- Subnet maps[7] $F$ and $G$ we realize as maps between the vector of priors $\beta_h$ and $\beta_{pr}$ for $B_h$ and $B_{pr}$ respectively:

$$F\colon\ [0,1]^{dim(\Gamma(B_h))} \to [0,1]^{dim(\Gamma(B_{pr}))}$$

$$G\colon\ [0,1]^{dim(\Gamma(B_{pr}))} \to [0,1]^{dim(\Gamma(B_h))}$$

  where we insist that the image of any encoding of a probability distribution encodes a probability distribution in the target (e.g. all primitives associated with a node sum to 1, etc.). $F$ and $G$ are controlled by hidden variables $f$ and $g$ respectively.

The reporter engages in what we will call the "translation procedure" which is as follows.

1. Pass a subgraph $\Gamma_q \subseteq \Gamma(B_h)$ specified by a question to $T$.
2. Endow $T(\Gamma_q)$ with its priors in $B_{pr}$ to get the vector $\beta_{pr}$.[8]
3. Pass $\beta_{pr}$ to $G$ to get a vector encoding updated priors in $B_h$
4. Pass $G(\beta_{pr})$ to $F$ to get a vector encoding updated priors in $B_{pr}$.
5. Pass $F(G(\beta_{pr}))$ to $G$ to update priors in $B_h$ a final time
6. Subject the updated $B_{pr}$ to prediction tests and compare the first and second updates of $B_h$.

After the translation procedure has occurred, we evaluate the performance of our maps and update their hidden variables according to gradient descent on three loss functions $L_T$, $L_F$, and $L_G$, all of which depend on $L_{pr}$.

## Defining $L_T$

$L_T$ should guide $T$ by penalizing it for associating a subgraph of $\gamma \subseteq \Gamma(B_h)$ with subgraphs of $\Gamma(B_{pr})$ that don't track with persistence. If $r$ is a variable tracking time, and $T_r$ is the predictor's version of the map $T$ at time $r$, then the loss function should reward $T_{r_1}(\gamma) \subseteq T_{r_2}(\gamma)$ when $r_1 < r_2$ and punish the opposite unless the predictive gains to be had are substantial. We put forward

$$L_T = L_{pr} + c \cdot exp(\#\ of\ persistence\ components - 1)$$

where $c$ should be determined from experimentation.

[7] We acknowledge the possible necessity of mapping into something containing $B$ as a substructure as well as the possibility of a counterexample if the relationship between these maps is underspecified.

[8] We interpret this to mean letting the boolean matrix for the subgraph act on the vector of priors to isolate all priors relevant to the computation. This is not really different from a look-up table.

## Defining $L_F$ and $L_G$

Assuming $B_{pr}$ has an effective strategy, $L_F$ should prevent $F$ from doing nothing[9] to $B_{pr}$ by punishing $F$ for changing $B_{pr}$ in a manner that doesn't scale with how much $B_h$ changes between the first and second updates, normalizing individual changes of priors in $B_{pr}$ according to frequency of node use.[10] Let $\phi$ be a vector assigning each prior associated with a node a value in $[0, 1]$ determined by the node's use-frequency.[11] If $F$ changes the vector of priors in $B_{pr}$ from $w$ to $w'$ and if $v$ and $v'$ are vectors representing the first and second updates of $B_h$, we encourage[12]

$$||\phi \cdot (w - w')||_p \geq ||v - v'||_p$$

by passing the updated difference $Upd_F = ||v - v'||_p - ||\phi \cdot (w - w')||_p$ to $L_f$.

We also introduce an adversarial relationship between $F$ and $G$ by allowing them to audit one another. We'll associate to $G$ an auditing map $Aud_G$ which performs the following:

1. Store a copy of the vector of priors $w$ for $B_{pr}$.
2. Generate some vector of priors $w'$ and pass it to $G$.
3. Pass $G(w')$ to $F$ to get a new vector of priors $w''$
4. Compute $diff = ||w - w'||_p - ||w - w''||_p$ and store it.
5. Repeat steps 1 - 4 while recording $diff_{max}$, the maximum $diff$ across the first $floor(N/e)$ repetitions.[13]
6. Now repeat steps 1 - 4 until $diff > diff_{max}$ and return $diff$.

We expect $diff$ to be small if $F$ is actually performing a faithful translation of priors. By auditing multiple times, $G$ is able to probe $F$ for inconsistencies in its translations and select the most egregious translation disparity as a reward. We have $F$ audit $G$ in a similar way with its own map $Aut_F$, but acknowledge the that differences in the sizes of $B_{pr}$ and $B_h$ may require these auditing functions to be different. Regardless, once the translation procedure has been completed and the three values $Upd_F$, $Aud_F$ , and $Aud_G$ have all been computed, we evaluate the performance of $F$ with the function

$$L_F = L_{pr} + Upd_F - Aud_F + Aud_G^{\chi_F}$$

Where $\chi_F$ counts the number of iterations in a row $Aud_G$ has exceeded some threshold we expect to decrease as time goes on. This asymmetry in the auditing component of the loss is intended to prevent $F$ and $G$ from developing a collaborative strategy where they exhibit detectable inconsistencies in a mutually beneficial way. We think $L_G$ should be defined very similarly.

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[9] Either by literally doing nothing or by only updating nodes that aren't relevant for good predictions.

[10] Returning to the teacher-student analogy, this condition is intended to prevent student responses that don't represent real understanding, like "Whatever the teacher said" or "It is what it is."

[11] That is, what percentage of the time the node is actually used in computations.

[12] The dot is the dot product and p is a real number determined via experimentation

[13] This is an implementation of the optimal strategy in the so-called secretary problem. *N* is some predetermined "maximum" number of trials. We think this strategy is a useful computational shortcut here since it performs very well (converging on largest choice over a third of the time, no matter the size of *N*) while always taking fewer steps than computing the true maximum.

## Counterexample: bad domain-relevant epistemology may not generically yield poor predictions in the short term

We specify in the proposal that posed questions be domain-relevant to deal with trivial counterexamples,[14] but we can conceive of domain-relevant questions for which false beliefs have no apparent impact on predictive effectiveness. There may be lots of ways to learn effective prediction with absurd epistemology; this could be bad in the long run. A silly illustration: during training the ehSV somehow begins to believe ghosts live in the vault. Superstition at a basic level is likely benign and may coincide with positive strategic adaptations found elsewhere, so it persists. Epistemic rot[15] settles in as superstitions aggregate and the ehSV begins devoting a substantial percentage of its computational efforts to thwarting paranormal robbery while still effectively predicting realistic scenarios.

## Strategy: Occam's razor (e.g. penalize complexity, computation time, downstream variables)

As mentioned in the proposal outline, the strategies "penalize complexity" (pp.38-39) and "penalize computation time" (pp.40-1) could be adapted so that the ehSV is incentivized to take Occam's razor to its belief system by pruning $B_{pr}$. One possibility might be to have the predictor $B_{pr}$ occasionally checks to see if certain low-traffic subnets are necessary for good inference by cutting them out and weighing the outcome in predictive tests on the labeled data.

## Counterexample: $G$ could lie and $F$ could "un-lie"

This issue arises if the relationship between $G$ and $F$ is underspecified. The human simulator could appear if the reporter constructs $G$ such that it distorts the representation of $\beta_{pr}$ in $B_h$ but builds an $F$ which undoes this distortion with minimal negative impact on the configuration of $B_{pr}$. Left unresolved, this offers a way for the human simulator to arise when the overhead of simulating the human is lower than faithfully translating $B_{pr}$.

## Strategy: Impose strong dependence relations between $F$ and $G$

We have attempted to impose strong dependencies between $F$ and $G$ and forbid obvious collaborative strategies, but readily acknowledge that more work needs to be done to ensure the proper adversarial relationship is attained and that the equilibrium state is reasonably accurate translation between $B_h$ and $B_{pr}$.

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[14] A child asks ehSV what the moon is made out of. Since this has no bearing on whether it can protect the diamond, ehSV can respond with "Cheese" which the child believes and the relevant updating of priors in $B_r$ has no impact whatsoever on ehSV's prediction strategy.

[15] The predictor acquires a false belief having little to no predictive impact (perhaps coinciding with an acquisition of good predictive strategy). A bizarre belief-system gradually grows without being pruned by the loss function until the predictor develops unacceptably inefficient strategies but is now in "too deep" to make better decisions. An analogy might be someone with severe psychosis who, despite receiving negative environmental feedback from their actions, is incapable of making better decisions.